

Maxson Stephen Mathew

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SUMMARY

Data Scientist and AI Engineer with a Master of Data Science (University of Melbourne) and hands-on data engineering experience spanning API integration, orchestration, and cloud-hosted pipelines. At John Holland, led a team delivering a 91%-accurate employee turnover prediction model across 600,000+ HR records. Independently built and shipped The Basis Point, an end-to-end pipeline integrating government and live banking APIs with dbt, Airflow, Docker, and CI/CD — plus a RAG-based invoice extractor (90% accuracy) and a \$109B+ retail analytics dashboard. Comfortable across the full stack from raw API extraction through orchestration to stakeholder-facing dashboards.

SKILLS

Languages & Analytics: Python | R | SQL | Statistical Analysis | Predictive Modelling | Feature Engineering.

Data Engineering: dbt | Apache Airflow | Docker | ETL/ELT Pipelines | PostgreSQL | Supabase | API Integration | CI/CD (GitHub Actions).

Machine Learning: Random Forest | Logistic Regression | KNN | Agent-Based Modelling | Social Network Analysis.

AI & LLM: LangChain | RAG Architecture | LLM Integration | Prompt Engineering | Semantic Extraction.

Business Intelligence: Power BI | DAX | Tableau | Excel | Streamlit | Data Storytelling.

Cloud Platforms: Microsoft Azure (Fabric, Lakehouse) | Google Cloud Platform (BigQuery) | Supabase.

Tools & Libraries: NetworkX | QGIS | PostGIS | pandas | scikit-learn.

EDUCATION

Professional Year Program (ICT)

May 2026 - Present

- Undertaking structured professional development to strengthen workplace readiness in the Australian ICT/data sector.

Master of Data Science

Feb 2023 - July 2025

The University of Melbourne, Parkville, Melbourne

- Delivered stakeholder-facing analytics projects translating complex, multi-system data into actionable organisational insights across technical and non-technical audiences.
- Developed expertise in data governance, spatial analytics, machine learning, and cloud-based BI platforms.

Bachelor of Technology – Computer Science

Aug 2017 - Jul 2021

Guru Gobind Singh Indraprastha University, Delhi, India

PROFESSIONAL EXPERIENCE

Data Science Intern

Feb 2024 - Oct 2024

John Holland Group, Melbourne, Australia

- Faced fragmented, inconsistent HR data across 600,000+ records (duplicates, supervisor mismatches, schema drift); resolved it by building repeatable data governance standards, enabling reliable downstream modelling.
- Needed to identify which employees were most influential to retention risk; built organisational network graphs across 10,000+ employees using Python/NetworkX, calculating centrality metrics to surface key influencers.
- Required a predictive view of turnover risk for workforce planning; led development of an Agent-Based Model combining centrality metrics with HR data, achieving 91% prediction accuracy that directly informed retention strategy.
- Needed a single technical point of contact between two organisations; served as sole stakeholder liaison between the University of Melbourne and John Holland, translating findings for senior decision-makers.

Assistant System Engineer (Data & Reporting)

Aug 2021 – Dec 2022

Tata Consultancy Services, Bengaluru, India

- Faced slow incident diagnosis on production banking systems for a regulated client (Deutsche Bank Belgium); introduced evidence-based root-cause analysis, cutting mean time to resolution by 95%.
- Needed to maintain data integrity across production banking systems under compliance requirements; implemented structured validation and monitoring, contributing to 99.5% application uptime.
- Required technical findings to be actionable for non-technical stakeholders; translated data insights into operational improvements across cross-functional teams.

Floor Staff & Supervisor

April 2023 - Present

Toscano's of Kew, Melbourne, Australia

- Managed operational pressure during closing shifts alongside full-time postgraduate study; delegated tasks across the team to consistently meet service standards.
- Onboarded new staff needing rapid ramp-up; trained new hires and delivered consultative customer service during peak periods.

PROJECTS

The Basis Point — Live Bank Rate & Inflation Analytics Pipeline

June 2026 - July 2026

Personal Portfolio Project | Python | dbt | Apache Airflow | Docker | PostgreSQL | Streamlit | GitHub Actions CI/CD

- No free tool existed to compare how Australian banks price loans and savings against current inflation; built a fully automated pipeline pulling live data from government and banking sources to answer this in real time.
- Banking data sources were unreliable and inconsistently structured across providers; engineered a self-correcting extraction process that adapts automatically instead of breaking when a source changes, reducing ongoing maintenance to near zero.
- Data accuracy was critical to any insight being trustworthy; built automated quality checks into the pipeline that caught and fixed a real data error before it ever reached the final report.
- Raw data alone isn't decision-ready; delivered an interactive dashboard that automatically surfaces which bank offers the most competitive rate in each category, turning a manual research task into a 5-second answer.

Victoria Retail Turnover Analysis (2023–2025)

Jan 2026 - Feb 2026

Personal Portfolio Project | Power BI | DAX | Excel | ABS Data | Business Intelligence

- Raw ABS retail data needed to be turned into an executive-usable view; designed a 5-page Power BI dashboard analysing \$109B+ in Victorian retail turnover across 30 months and 10+ categories.
- Needed forward-looking, not just historical, insight; authored 8 custom DAX measures including forecasting logic, surfacing a 2.9% annual growth trend and a \$12.1B December 2024 peak.

AI-Powered Invoice Data Extractor

Dec 2025 - Jan 2026

Personal Portfolio Project | RAG | LangChain | LLM | Streamlit | Python

- Manual invoice data entry was slow and error-prone with no format-specific rules available; built a RAG pipeline (LangChain) to extract and structure data from unstructured invoice PDFs, reaching 90% extraction accuracy end-to-end.
- Needed the solution to be usable without local setup; deployed on Streamlit Cloud, estimated to cut manual data entry by 80-90%.

AMI Classification & Troponin Prediction

July 2024 - Nov 2024

Academic Project University of Melbourne | Python | SQL | Machine Learning | Google Big Query

- Clinical predictors of AMI needed to be identified from fragmented ICU data; extracted a clinical cohort via multi-table SQL joins across admissions, lab events, and chart data in BigQuery.
- Needed a validated, best-performing predictive model; evaluated KNN, Logistic Regression, and Random Forest; Random Forest achieved 94% accuracy and 0.95 ROC-AUC, confirming troponin as the strongest AMI predictor.

PROFESSIONAL CERTIFICATIONS

- Claude Code in Action (Anthropic)
- CITI Program - Human Subjects Research (Clinical Data Governance & Ethics)
- Databricks Fundamentals Accreditation
- Databricks AI Agents Fundamentals Accreditation

INTERESTS & COMMUNITY

Active member of one of three Chenda percussion groups in Melbourne, performing at cultural events across Victoria. Interests include cinema, emerging AI technology, and travel.